

ISEN/DAEN 427 Notes for Lecture 4

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MSE: Bias–Variance Decomposition



Mean Squared Error (MSE)

Definition

For an estimator $\hat{\theta}$ of a true parameter θ ,

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

- Measures average squared error between estimates and truth.
- Will decompose into **variance** and **bias**².

Step 1: Add and subtract the mean of the estimator

$$\text{MSE}(\hat{\theta}) = \mathbb{E} \left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta \right)^2 \right]$$

Let

$$A = \hat{\theta} - \mathbb{E}[\hat{\theta}], \quad B = \mathbb{E}[\hat{\theta}] - \theta$$

Step 2: Expand the square

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= \mathbb{E}[(A + B)^2] \\ &= \mathbb{E}[A^2] + 2\mathbb{E}[AB] + \mathbb{E}[B^2]\end{aligned}$$

- A is a zero-mean random variable: $\mathbb{E}[A] = 0$
- B is a constant (non-random) term: the **bias** $\text{Bias}(\hat{\theta})$

Step 3: Take expectations term-by-term

$$\mathbb{E}[A^2] = \text{Var}(\hat{\theta}),$$

$$2 \mathbb{E}[AB] = 2 B \mathbb{E}[A] = 0,$$

$$\mathbb{E}[B^2] = B^2 = (\text{Bias}(\hat{\theta}))^2.$$

Result: Bias–Variance Decomposition

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.$$

- Variance: *inconsistency* / spread across samples
- Bias²: *systematic error* from truth (always positive due to squaring)

Example: estimating a coin's probability of heads

True probability $p = 0.5$.

Estimator A: sample mean of $n = 4$ flips

Unbiased so Bias = 0.

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{0.5 \cdot 0.5}{4} = 0.0625.$$

$$\text{MSE} = \text{Var} + \text{Bias}^2 = \boxed{0.0625}.$$

Example (continued)

Estimator B: constant guess $\hat{p} = 0.6$

$$\text{Bias} = 0.6 - 0.5 = 0.1, \quad \text{Var} = 0$$

$$\text{MSE} = 0 + 0.1^2 = \boxed{0.01}$$

Estimator C: constant guess $\hat{p} = 0.8$

$$\text{Bias} = 0.8 - 0.5 = 0.3, \quad \text{Var} = 0$$

$$\text{MSE} = 0 + 0.3^2 = \boxed{0.09}$$

- MSE naturally separates into **variance** and **bias**² by expanding $\mathbb{E}[(\hat{\theta} - \theta)^2]$.
- Bias is squared so positive/negative systematic errors are penalized equally and grow quadratically with magnitude.
- A slightly biased, low-variance estimator (like the constant 0.6 here) can have lower MSE than an unbiased, higher-variance one.